

Hotel Booking Analysis

Research Question

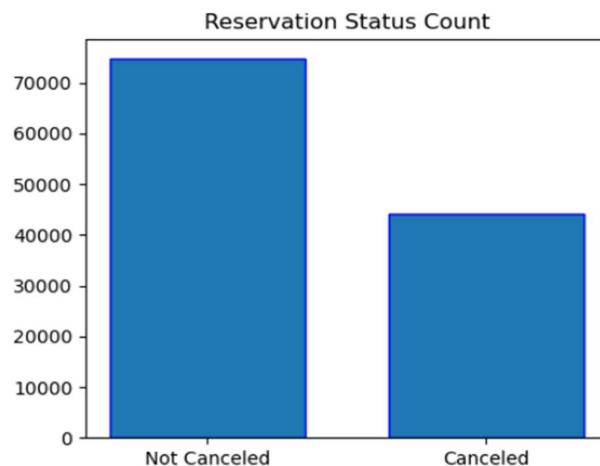
1. What are the variables that affect hotel reservation cancellations?
2. How can we make hotel reservation cancellations better?
3. How will hotels be assisted in making pricing and promotional decisions?

Hypothesis

1. More cancellation occurs when the prices are higher.
2. When there is a long waiting list, customers tend to cancel more frequently.
3. The majority of the clients are coming from offline travel agents to make their reservations.

Analysis and Findings

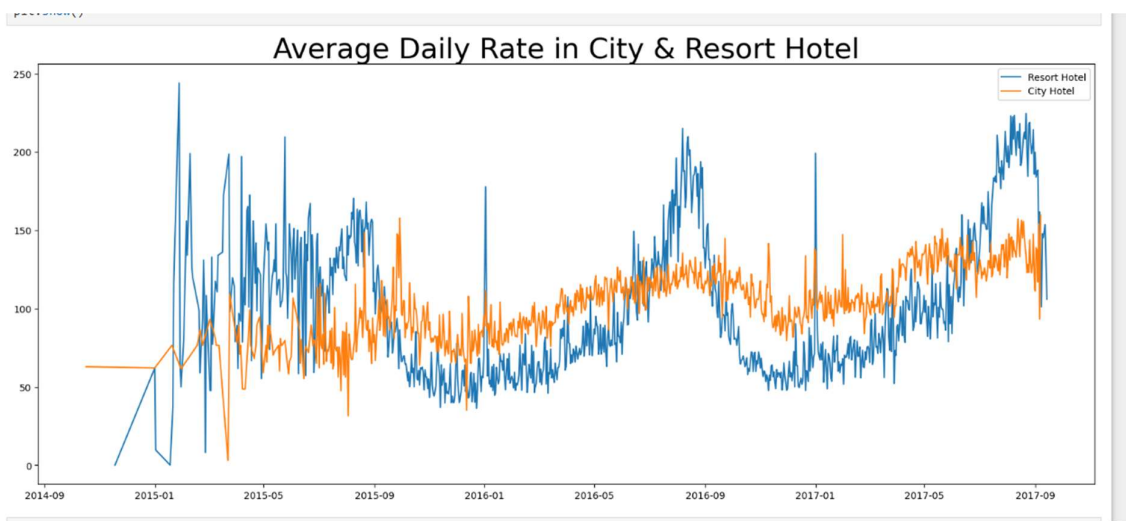
```
is_canceled
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1    0.371347
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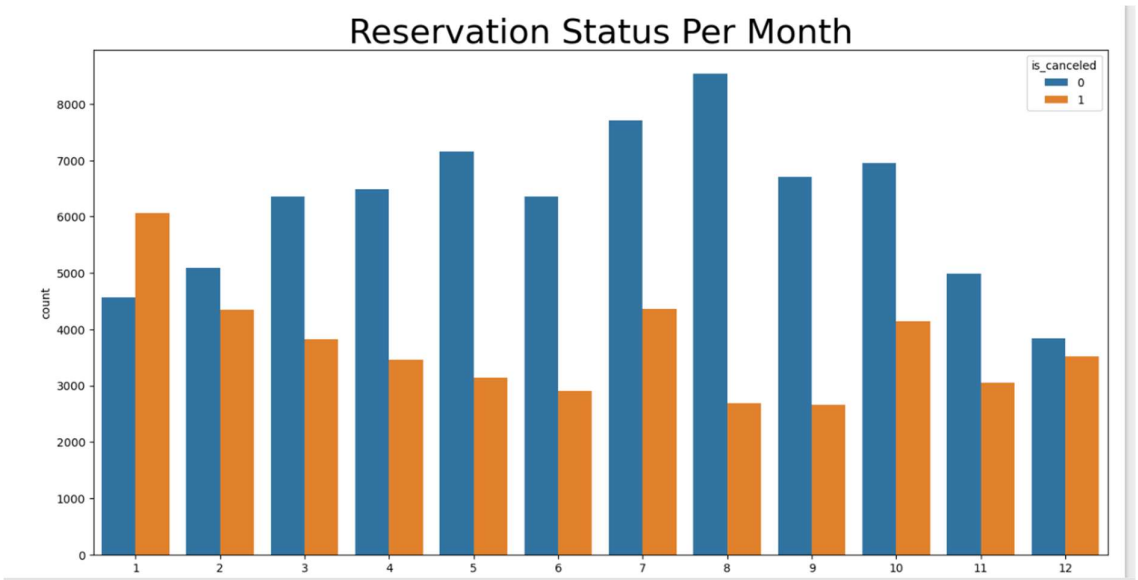
The accompanying bar graph shows the percentage of reservations that are canceled and those that are not. It is obvious that there are still a significant number of reservations that have not been canceled. There are still 37% of clients who canceled their reservations , which has a significant impact on hotel's earning.



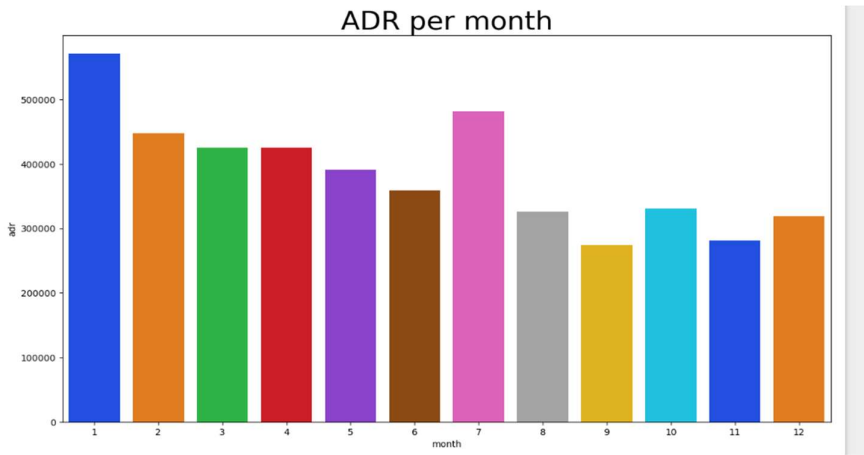
In comparison to resort hotels, city hotels have more bookings. It's possible that resort hotels are more expensive than those in cities.



The line graph above shows that , on certain days, the average daily rate for a city hotel is less than resort hotel ,and on other days ,it is even less. It goes without saying that weekends and holidays may see a rise in resort hotel rates.



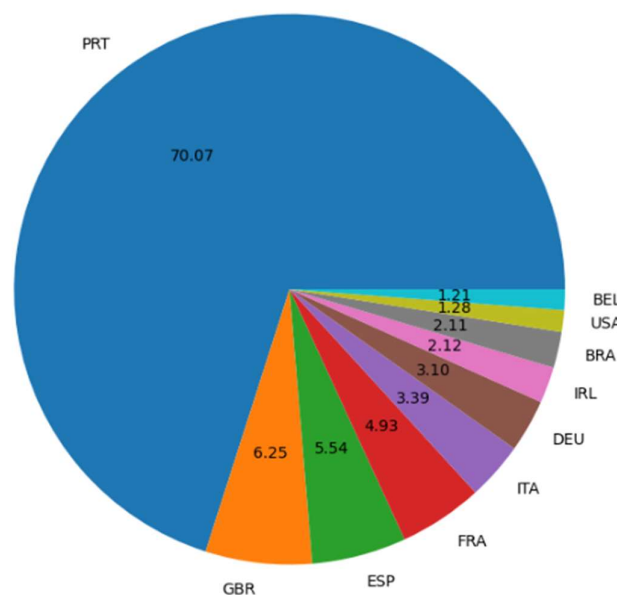
We have developed the grouped bar graph to analyze the months with the highest and lowest reservation levels according to reservation status. As can be seen, the number of confirmed reservations are largest in the month of August , whereas January is the month with the most canceled reservations.



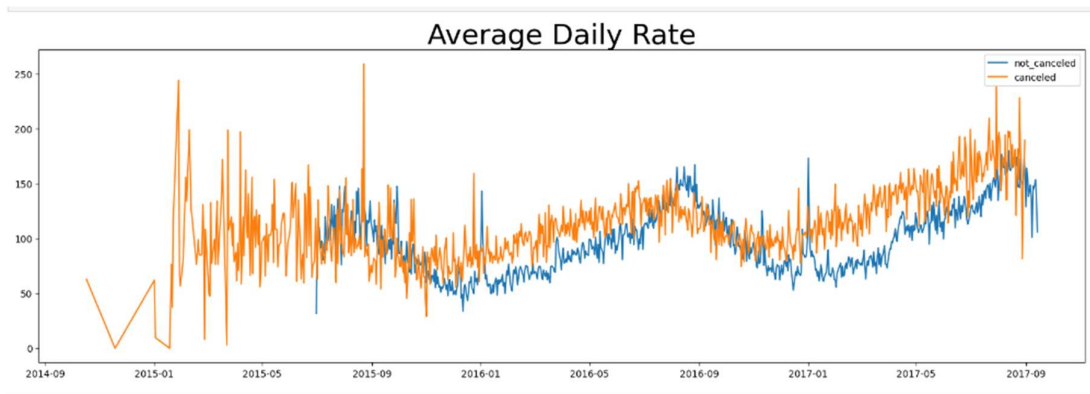
This bar graph demonstrates that cancellations are most common when prices are greatest and are least common when they are lowest. Therefore, the cost of the accommodation is solely responsible for the cancellation.

Now, let's see which country has the highest reservation canceled. The top country is Portugal with the highest number of cancellations.

Top 10 countries with reservation cancelation



Let's check the area from where guests are visiting the hotels and making reservations. Is it coming from Direct or Groups. Online or Offline Travel Agents? Around 46% of the clients come from online travel agencies, whereas 27% come from groups. Only 4% of clients book hotels directly by visiting them and making reservations.



As seen in the graph, reservations are canceled when the average daily rate is higher than when it is not canceled. It clearly prove all the above analysis , that the higher price leads to higher cancelations.

Suggestions

1. Cancellation rates rise as the price does. In order to prevent cancelations of reservations , hotel could work on their pricing strategies and try to lower the rates for specific hotels based on locations. They can also provide some discounts to the consumers.
2. As the ratio of cancelation and not cancelation of the resort hotel is higher in the resort hotel than the city hotels. So the hotels should provide a reasonable discount on the room prices on weekend or on holidays.
3. In the month of Januray, hotels can start campaigns or marketing with a reasonable amount to increase their revenue as the cancelation is the highest in this month.
4. They can also increase the quality of their hotels and their services mainly in Portugal to reduce the cancellation rate.